

YONAGO, Japan

WMO: 477440

Lat: 35.4347N

Lon: 133.3386E

Elev: 27

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.9	31.4	20.6	15.5	38.1	22.6	17.1	38.8	20.6	38.7	18.6	39.3	6.9	160	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.1	93.2	77.9	91.1	77.5	88.8	77.0	79.5	88.5	78.7	87.6	77.8	86.4	10.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	141.3	84.0	76.2	136.8	83.4	75.3	132.7	82.9	43.1	88.5	42.2	87.8	41.3	86.8	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.8	15.5	13.8	DB	26.3	97.4	2.5	1.7	24.5	98.6	23.0	99.6	21.6	100.6	19.7	101.9
			WB	24.7	80.9	2.4	1.0	23.0	81.6	21.6	82.2	20.2	82.8	18.5	83.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	40.3	41.2	46.6	55.7	65.0	71.5	79.8	81.9	73.7	63.8	54.1	44.9
	DBStd	15.16	4.50	5.13	6.00	6.37	5.30	4.77	5.00	4.31	5.23	5.07	5.50	5.26
	HDD50	904	302	251	141	18	0	0	0	0	0	0	21	171
	HDD65	3390	765	665	569	285	67	3	0	0	1	82	330	623
	CDD50	4554	2	6	37	190	464	646	924	989	712	428	143	13
	CDD65	1565	0	0	0	7	66	199	459	524	263	45	2	0
	CDH74	13613	0	0	2	52	374	1030	4479	5746	1754	172	4	0
	CDH80	4868	0	0	0	6	54	221	1728	2387	448	24	0	0
(14) Wind	WSAvg	6.2	6.4	6.4	6.8	6.6	6.4	5.7	5.9	6.0	5.8	5.7	5.9	6.3
(15) Precipitation	PrecAvg	73.2	5.9	5.4	4.9	4.6	4.8	7.6	9.3	5.7	8.9	5.3	5.1	5.0
	PrecMax	92.0	11.4	10.7	9.7	7.6	11.0	20.1	24.4	18.3	18.0	11.7	9.4	8.3
	PrecMin	48.7	2.0	1.4	2.1	1.7	1.8	2.2	0.3	0.3	1.1	1.6	1.9	1.7
	PrecStd	10.9	1.9	1.7	1.5	1.5	2.0	4.0	6.2	3.8	4.5	2.6	1.7	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.3	64.4	71.1	79.4	84.8	89.1	95.0	97.0	92.2	83.3	73.8	63.7
		MCWB	51.6	55.0	55.7	61.9	65.5	74.3	77.4	77.9	77.0	70.2	62.3	54.6
	2%	DB	53.6	57.3	65.1	74.6	81.2	85.3	92.6	94.1	88.1	78.3	69.3	59.2
		MCWB	46.2	49.6	53.0	59.8	63.7	72.1	77.5	78.2	76.1	68.3	60.1	51.5
	5%	DB	50.7	53.7	61.2	70.9	78.2	82.5	90.5	91.8	84.7	75.1	66.6	56.6
		MCWB	44.4	46.9	51.7	57.9	62.9	71.3	77.3	77.8	74.8	66.2	58.6	49.8
	10%	DB	48.2	50.5	57.8	67.2	75.3	79.7	88.2	89.7	82.0	72.7	64.1	54.0
		MCWB	42.5	44.7	49.9	56.4	62.2	70.0	76.8	77.6	73.3	64.2	57.0	48.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	56.4	58.5	66.7	71.0	77.0	80.0	81.0	79.1	73.1	65.2	57.3
		MCDB	57.4	63.1	66.3	73.3	77.6	84.9	89.3	90.2	86.7	80.2	69.9	61.0
	2%	WB	47.8	51.0	55.3	62.3	68.0	75.2	79.0	79.9	77.3	70.5	62.3	53.1
		MCDB	52.1	55.4	61.4	71.4	74.8	82.1	88.4	89.1	85.0	76.3	67.2	57.7
	5%	WB	45.1	48.0	52.9	59.9	66.0	73.5	78.2	79.1	76.1	68.1	60.1	50.7
		MCDB	49.6	52.5	59.0	67.7	73.9	79.7	87.3	88.1	83.0	72.9	65.2	55.5
	10%	WB	43.0	45.1	50.7	58.0	64.4	71.8	77.4	78.3	74.8	65.7	57.8	48.4
		MCDB	47.7	49.7	57.2	64.6	72.3	77.6	86.1	87.0	81.0	70.8	63.2	53.1
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	12.7	15.6	17.1	16.3	12.5	12.0	13.1	12.8	14.9	14.5	12.2
		MCDBR	16.1	19.8	22.4	23.1	21.5	16.6	15.6	15.9	15.6	17.5	18.2	17.1
		MCWBR	10.9	13.5	13.7	12.0	10.0	7.1	5.6	5.7	6.7	9.0	10.8	11.3
	5% WB	MCDBR	14.7	16.7	18.6	17.8	15.9	12.5	14.3	14.2	13.2	13.7	14.7	15.3
		MCWBR	11.0	12.7	12.8	10.4	8.9	5.8	5.6	5.7	6.0	7.8	10.1	11.4
(40) Clear-Sky Solar Irradiance	taub		0.441	0.476	0.545	0.564	0.561	0.579	0.576	0.540	0.496	0.457	0.438	0.432
	taud		1.989	1.929	1.795	1.806	1.868	1.900	1.968	2.066	2.139	2.141	2.094	2.035
	Ebn at Noon		227	235	230	235	238	233	233	239	243	240	227	220
	Edh at Noon		44	52	65	67	64	62	58	51	46	42	39	40
(44) All-Sky Solar Radiation	RadAvg		546	777	1138	1504	1660	1547	1532	1595	1235	1007	689	496
	RadStd		44	78	102	125	158	121	227	179	109	90	56	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page